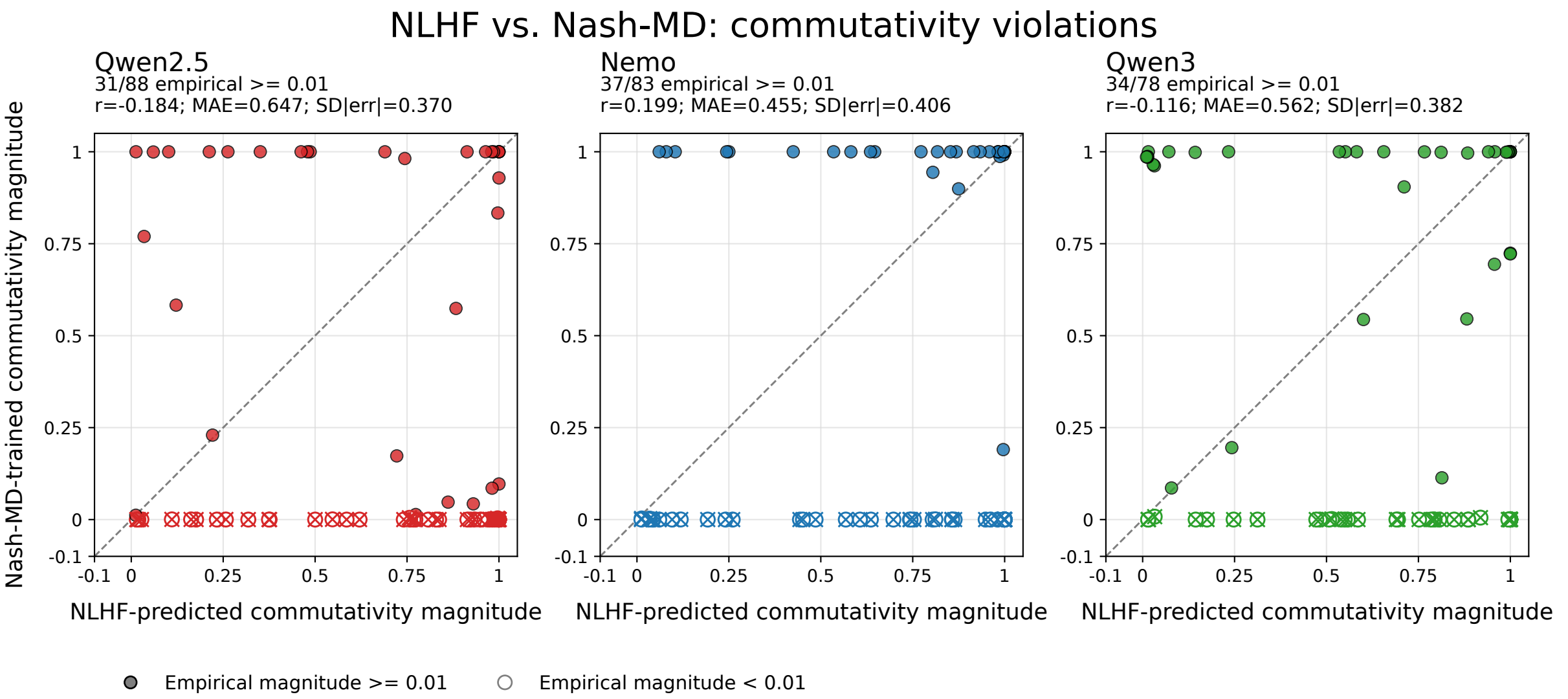

NLHF vs. Nash-MD: commutativity violations
Qwen2.5
31/88 empirical >= 0.01
r=-0.184; MAE=0.647; SD|err|=0.370
Nemo
37/83 empirical >= 0.01
r=0.199; MAE=0.455; SD|err|=0.406
Qwen3
34/78 empirical >= 0.01
r=-0.116; MAE=0.562; SD|err|=0.382
Nash-MD-trained commutativity magnitude
NLHF-predicted commutativity magnitude
NLHF-predicted commutativity magnitude
NLHF-predicted commutativity magnitude
-0.1
0
0.25
0.5
0.75
1
Empirical magnitude >= 0.01
Empirical magnitude < 0.01